

FACTORS AFFECTING STARTUP SURVIVAL: A SURVIVAL ANALYSIS OF VENTURE-FUNDED STARTUPS

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ABSTRACT

Startup failure remains a significant challenge within entrepreneurial ecosystems despite increasing access to venture capital funding. Understanding the factors associated with startup survival is therefore important for entrepreneurs, investors, and policymakers. This study investigates the determinants of startup survival using survival analysis techniques on venture-funded startups obtained from the Crunchbase Startup Dataset.

The original dataset contained 196,553 startups. After data cleaning and selection of venture-funded firms with valid founding dates, a final analytical dataset consisting of 21,725 startups was obtained. Startup failure was defined as closure of operations, while active, acquired, and initial public offering (IPO) firms were treated as right-censored observations. Survival time was measured as the number of days from founding until closure or censoring.

The study employed Kaplan–Meier survival estimation, log-rank testing, and Cox proportional hazards regression to evaluate differences in survival outcomes and identify significant predictors of startup failure. Kaplan–Meier analyses indicated that higher funding levels and a greater number of funding rounds were associated with improved survival probabilities. Significant survival differences were also observed across industry categories. Log-rank tests confirmed statistically significant differences in survival experiences among funding levels, funding rounds, and industries. Cox proportional hazards modelling identified funding levels, funding rounds, milestones, and relationship networks as statistically significant predictors of startup survival.

The findings demonstrate that financial resources and network-related characteristics play important roles in startup longevity. The study contributes to the growing literature on entrepreneurial survival and highlights the usefulness of survival analysis methods in examining startup outcomes.

1. *INTRODUCTION*

Entrepreneurship plays an important role in economic development, technological innovation, and job creation. Startups contribute significantly to modern economies by introducing new products, services, and business models. However, startup failure rates remain high, making the study of survival determinants an important area of research.

Venture-funded startups represent a particularly interesting segment of the entrepreneurial landscape. These firms often receive external funding to support growth and innovation but still face considerable uncertainty regarding long-term survival. Investors, founders, and policymakers seek to understand which characteristics are associated with startup success and failure.

Traditional statistical approaches often focus on binary outcomes such as success or failure. While useful, such methods may ignore the timing of events and the presence of incomplete observations. Survival analysis provides a suitable framework for studying startup longevity because it explicitly accounts for time-to-event data and censored observations.

This project applies survival analysis techniques to examine factors affecting startup survival among venture-funded firms. Using data obtained from the Crunchbase Startup Dataset, the study investigates how funding characteristics and organizational attributes influence the risk of startup failure over time.

The research employs Kaplan–Meier survival estimation to examine survival patterns, log-rank tests to compare survival distributions among groups, and Cox proportional hazards modelling to identify significant predictors of startup failure. Together, these methods provide a comprehensive assessment of startup survival dynamics.

2. *OBJECTIVES OF THE STUDY*

The primary objectives of this study are:

1. To analyse survival patterns among venture-funded startups.
2. To estimate startup survival probabilities using Kaplan–Meier survival analysis.
3. To compare survival experiences across funding levels, funding rounds, and industry categories.
4. To evaluate statistical differences in survival distributions using log-rank tests.
5. To identify significant predictors of startup failure using the Cox proportional hazards model.
6. To assess the influence of funding, milestones, and relationship networks on startup survival.
7. To demonstrate the applicability of survival analysis techniques in entrepreneurial research.
8. DATASET DESCRIPTION

The study utilised the Crunchbase Startup Dataset obtained from Kaggle. The dataset contains information on startups, funding activities, industry classifications, and company outcomes.

The original dataset consisted of 196,553 startup observations. Data cleaning and filtering procedures were conducted to retain only venture-funded startups with valid founding dates. Following preprocessing, the final analytical dataset contained 21,725 startups.

The variables used in the analysis included:

- status
- founded_at
- closed_at
- funding_total_usd
- funding_rounds
- category_code
- country_code
- milestones
- relationships
- survival_time_days
- event

Startup survival time was calculated from the founding date to either the closure date or censoring date. Closure represented the failure event of interest.

Table 1. Dataset Summary Statistics

Metric	Value
Initial Sample Size	196,553
Final Sample Size	21,725
Failure Events	1,301
Censored Observations	20,424
Mean Funding (USD)	\$15.2 Million
Median Funding (USD)	\$2.5 Million
Maximum Funding (USD)	\$5.7 Billion

Funding statistics for the final venture-funded dataset indicated substantial variability among startups. The mean funding amount was approximately \$15.2 million, while the median funding amount was approximately \$2.5 million. The maximum observed funding exceeded \$5.7 billion.

The most frequently represented industry categories included:

- Software (3,529 startups)
- Web (2,000 startups)
- Biotech (1,969 startups)
- Mobile (1,576 startups)
- Enterprise (1,283 startups)
- E-commerce (1,152 startups)

These figures demonstrate the diversity of industries represented within the venture-funded startup ecosystem.

4. METHODOLOGY

4.1 Data Cleaning

The original dataset underwent preprocessing before analysis. Startups without valid founding dates were removed because survival time calculations require a known starting point. Survival-related variables were subsequently created using startup status, founding dates, and closure dates.

For Cox regression analysis, complete-case analysis was employed. Observations with missing predictor values were excluded from the modelling stage.

4.2 Survival Time Definition

Survival time was defined as the number of days between a startup's founding date and either its closure date or censoring date.

This definition enabled measurement of the duration for which each startup remained operational.

4.3 Event Definition

The event of interest was startup failure.

Failure was defined as a startup having a status classified as "closed." Startups experiencing closure were considered event observations in the survival analysis framework.

4.4 Censoring Strategy

Not all startups experienced failure during the observation period. Therefore, right-censoring was incorporated into the analysis.

Startups classified as active, acquired, or IPO were treated as censored observations because failure had not been observed for these firms during the available observation period.

The final dataset contained:

- Failure events: 1,301
- Censored observations: 20,424

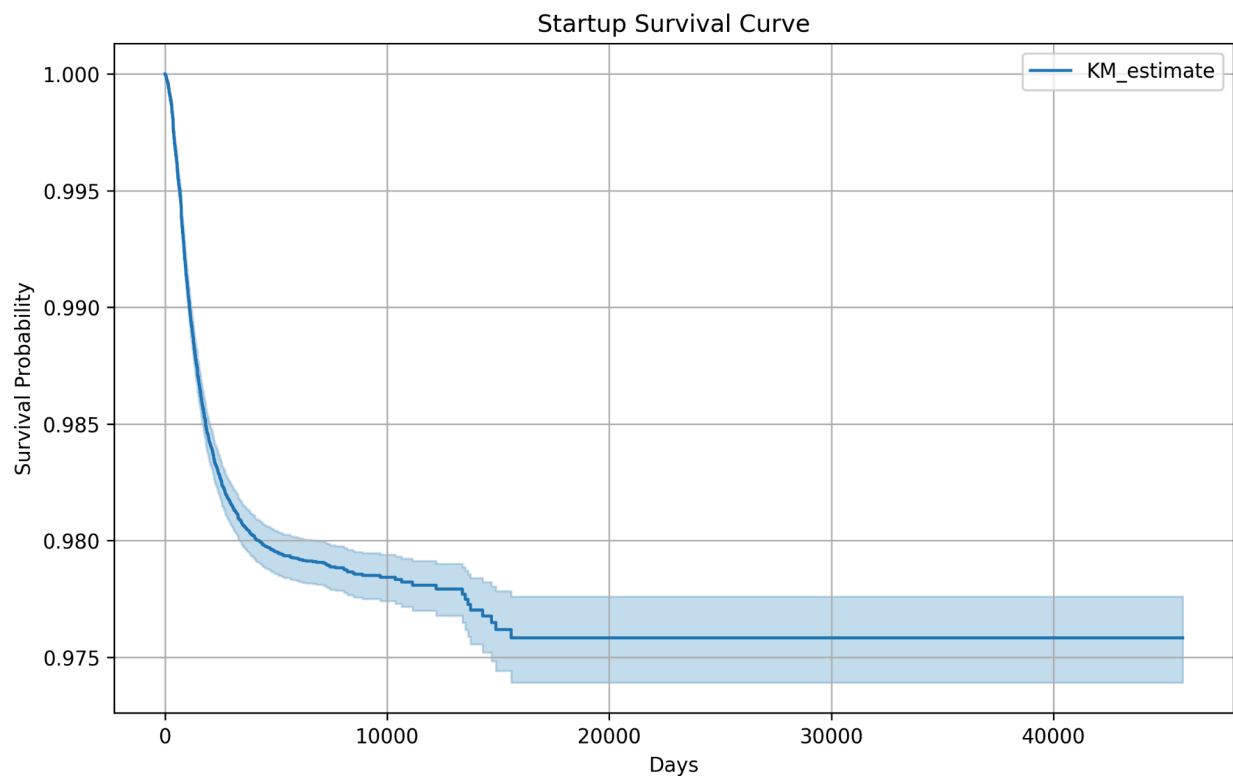
4.5 Kaplan–Meier Estimation

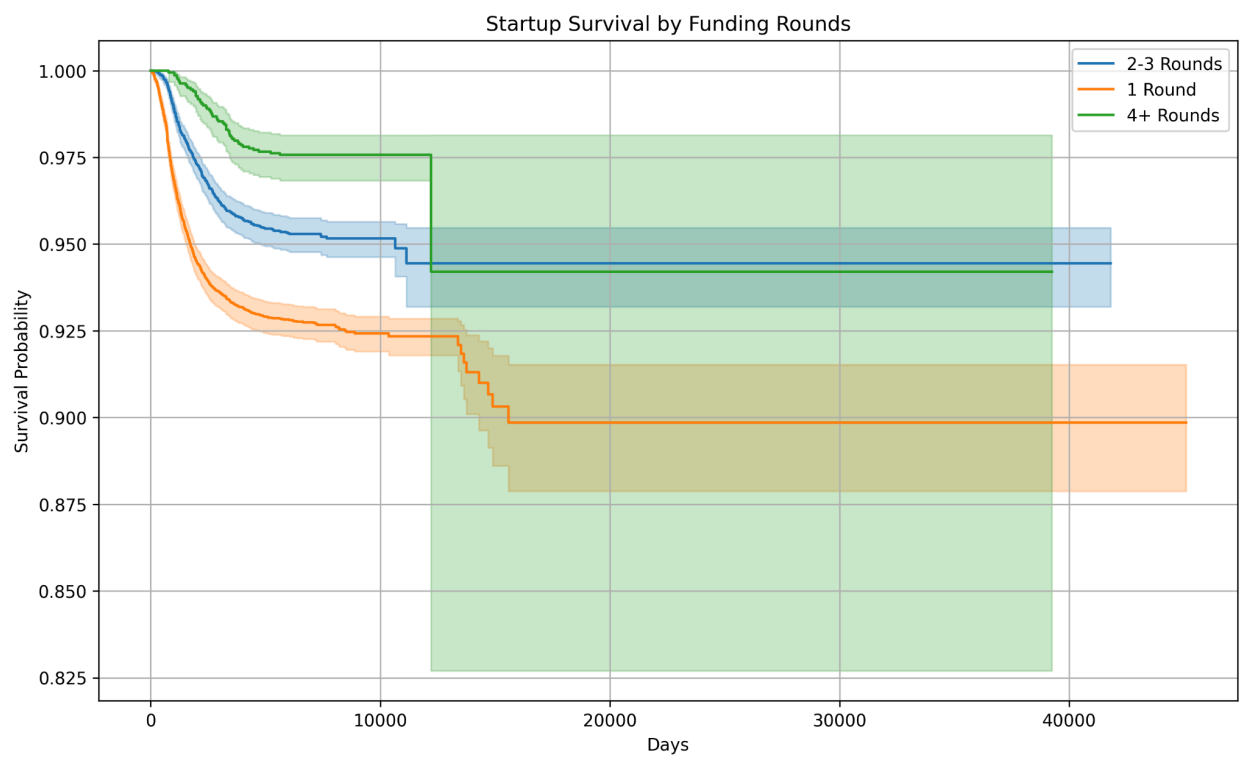
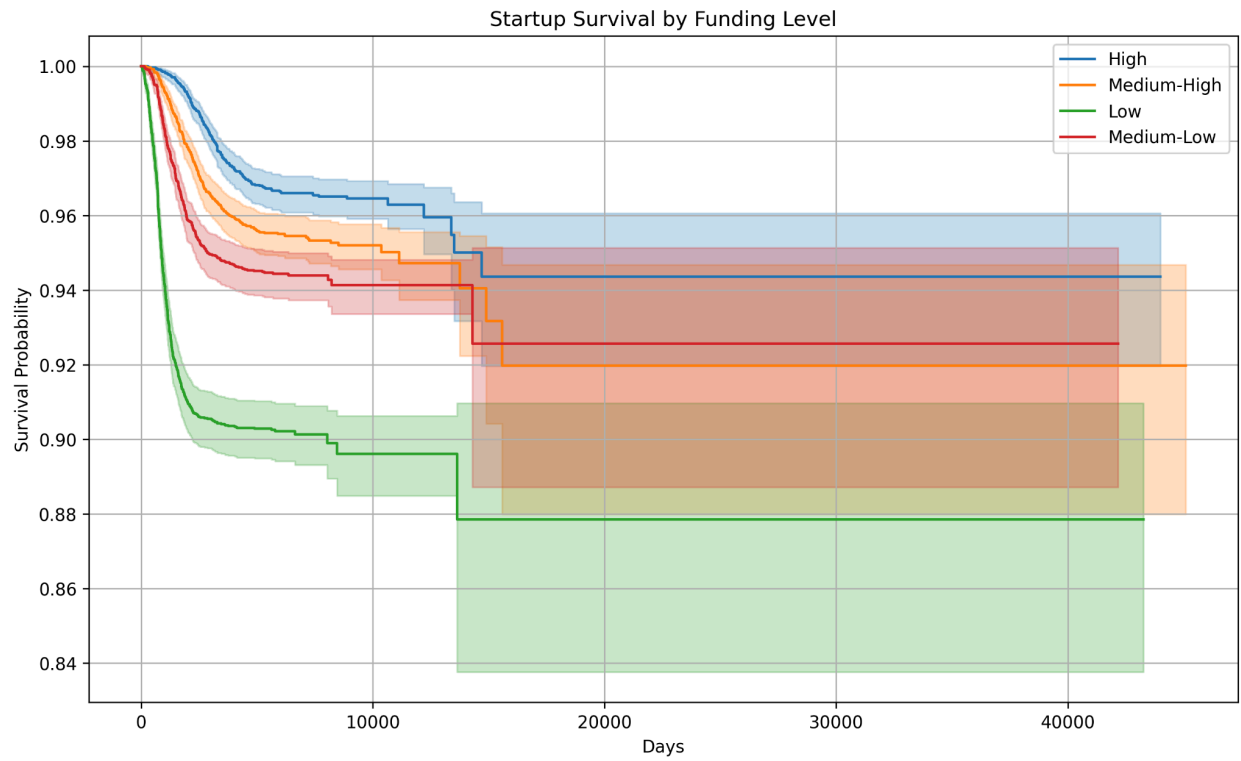
Kaplan–Meier estimation was used to estimate survival probabilities over time.

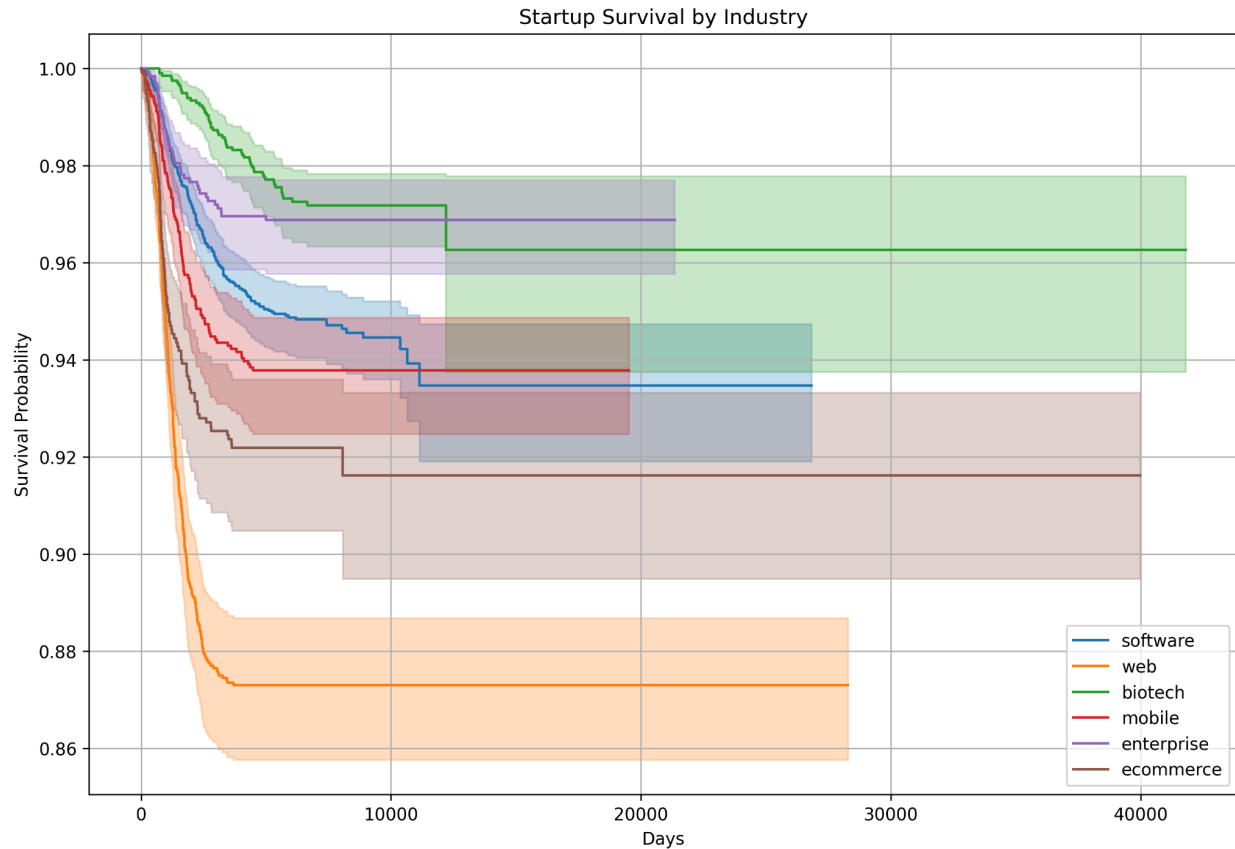
The Kaplan–Meier method provides a non-parametric estimate of the survival function and is particularly suitable for datasets containing censored observations.

Overall survival patterns were examined, followed by subgroup analyses based on:

- Funding level
- Funding rounds
- Industry category







4.6 Log-Rank Test

The log-rank test was used to compare survival distributions across different groups.

This test evaluates whether observed survival differences between groups are statistically significant across the entire follow-up period.

Separate log-rank tests were performed for funding levels, funding rounds, and industry categories.

4.7 Cox Proportional Hazards Model

To identify predictors of startup survival, a Cox proportional hazards regression model was estimated.

The model included the following predictors:

- log_funding
- funding_rounds
- milestones
- relationships

The Cox model evaluates how each predictor influences the hazard, or instantaneous risk, of startup failure while accounting for differing survival times among firms.

5. *RESULTS*

5.1 *Descriptive Findings*

The final venture-funded dataset consisted of 21,725 startups.

Among these firms, only 1,301 experienced closure during the observation period, while 20,424 observations were censored. This distribution suggests that most startups either remained operational or experienced outcomes other than failure.

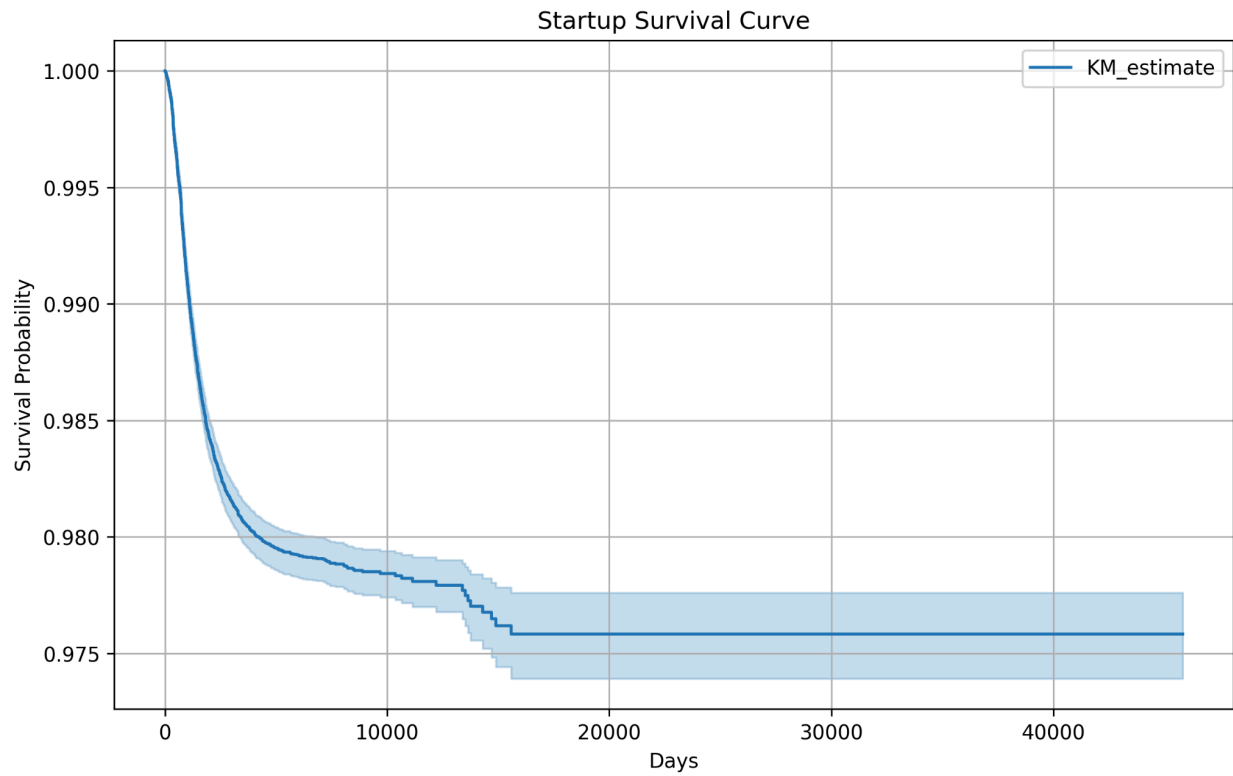
Funding amounts varied substantially across firms, indicating considerable heterogeneity in access to venture capital resources. Industry representation was also diverse, with software-related ventures constituting the largest category.

The descriptive findings indicate substantial variation in funding levels, funding rounds, milestones, and relationship networks among startups.

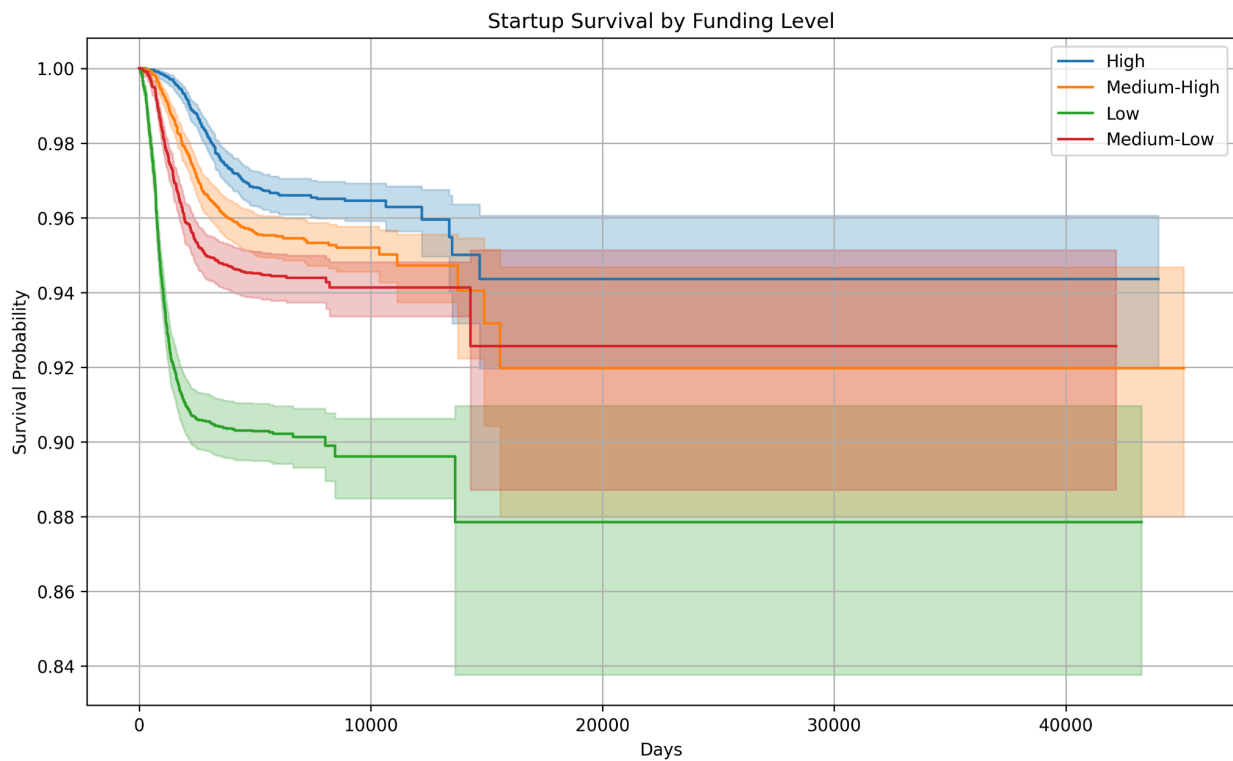
5.2 *Kaplan–Meier Results*

Kaplan–Meier survival analysis revealed generally high survival probabilities throughout the observation period.

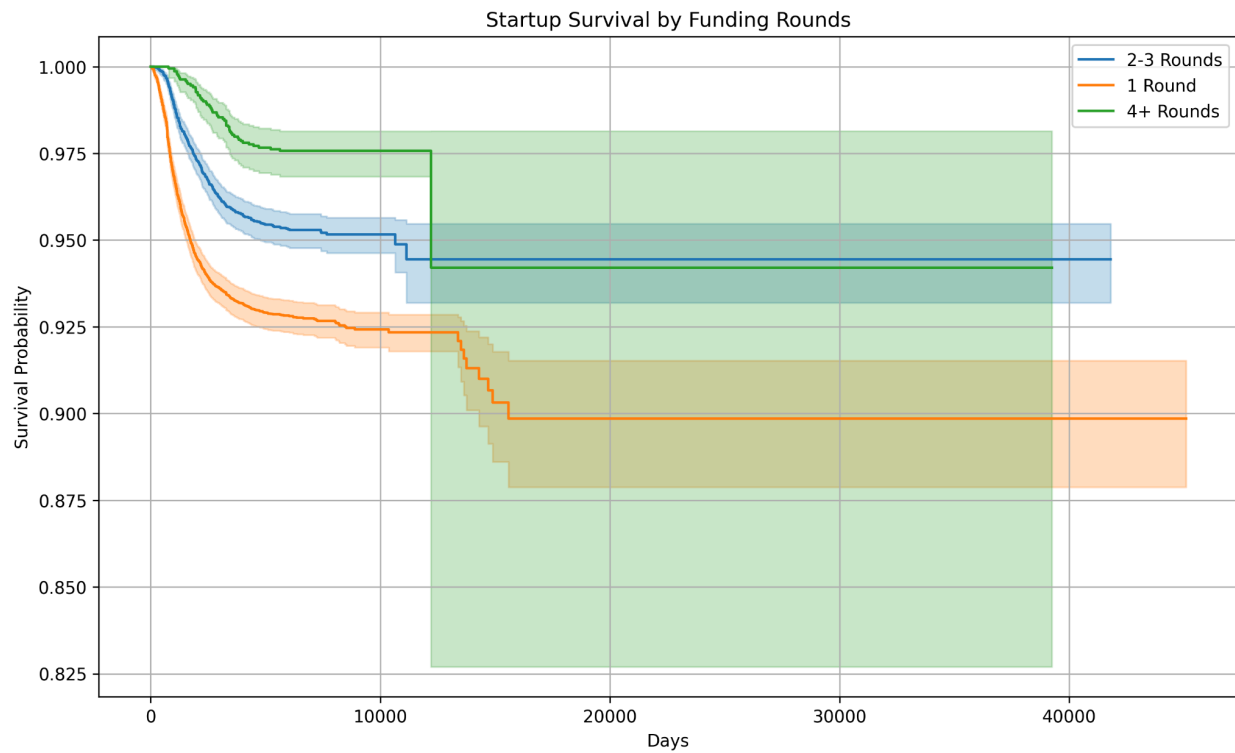
The overall survival curve showed gradual declines corresponding to observed startup failures rather than abrupt drops.



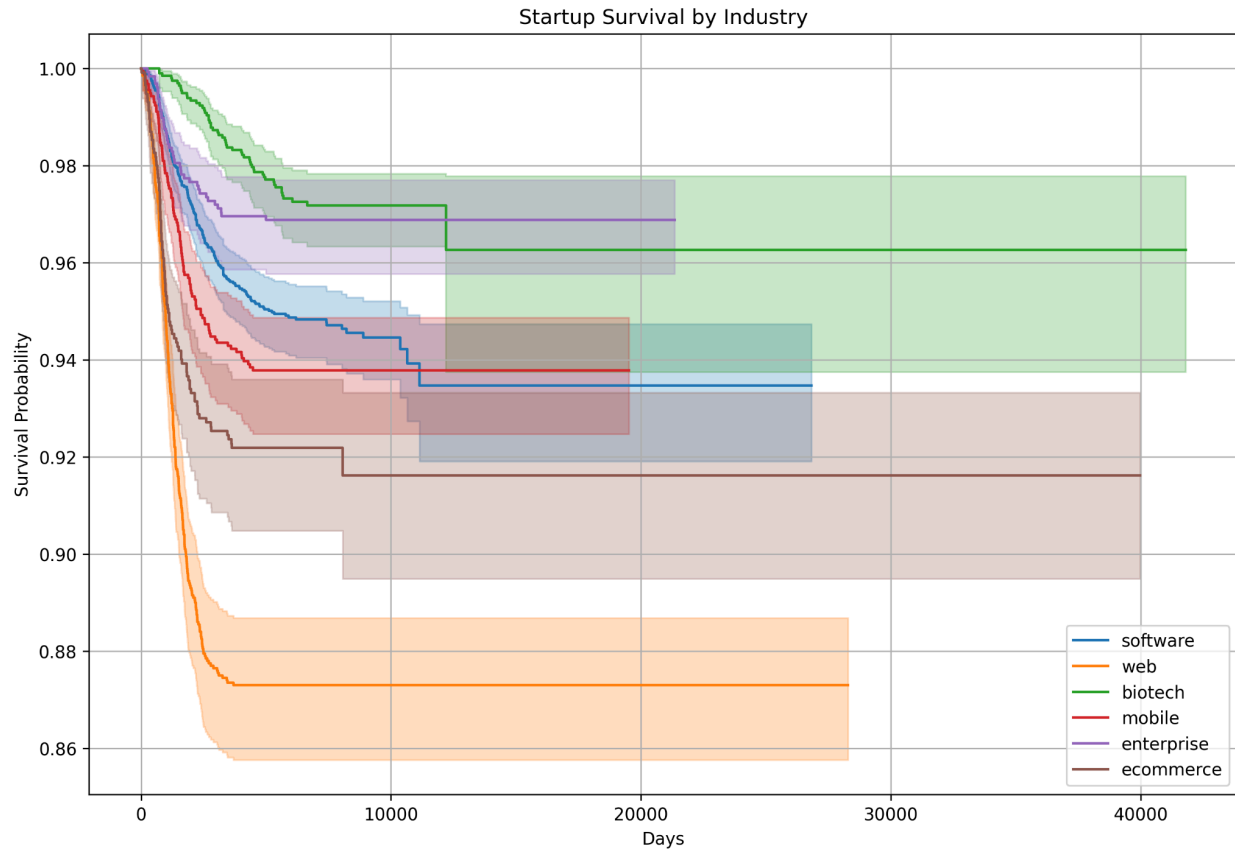
Survival curves stratified by funding level demonstrated that startups receiving higher funding generally exhibited better survival prospects.



Similarly, startups that participated in a greater number of funding rounds tended to survive longer.



Differences were also observed among industry categories, suggesting that startup survival experiences vary across sectors.



Overall, Kaplan–Meier analyses indicated meaningful differences in survival outcomes across startup characteristics.

5.3 Log-Rank Results

The log-rank test results supported the observed differences in survival curves.

For funding levels, the log-rank test statistic was 258.103 with a p-value of 1.1565×10^{-55} , indicating statistically significant differences in survival distributions across funding categories.

For funding rounds, the test statistic was 109.907 with a p-value of 1.3614×10^{-24} , suggesting significant differences in survival experiences among startups receiving varying numbers of funding rounds.

Industry categories also exhibited significant survival differences, with a log-rank statistic of 227.726 and a p-value of 3.2852×10^{-47} .

These results indicate that funding characteristics and industry classification are associated with differing survival outcomes.

5.4 Cox Model Results

The Cox proportional hazards model was estimated using 13,161 observations and 717 observed failure events.

The model achieved a concordance value of 0.68 and a log-likelihood ratio test statistic of 307.73.

Table 2. Cox Proportional Hazards Model Results

Variable	Hazard Ratio	95% CI	p-value
Log Funding	0.87	0.84 - 0.90	<0.005
Funding Rounds	0.91	0.83 - 0.99	0.03
Milestones	1.25	1.13 - 1.37	<0.005
Relationships	0.90	0.88 - 0.92	<0.005

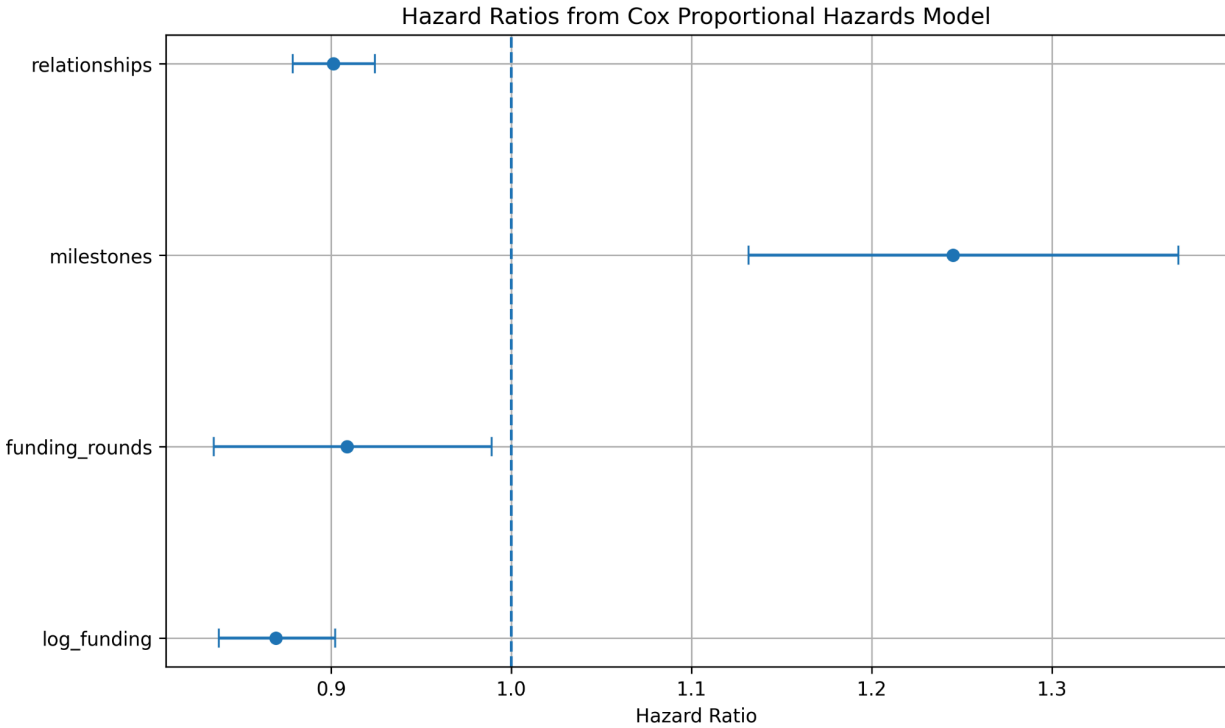
The estimated hazard ratios were:

- log_funding: HR = 0.87 (95% CI: 0.84–0.90)
- funding_rounds: HR = 0.91 (95% CI: 0.83–0.99)
- milestones: HR = 1.25 (95% CI: 1.13–1.37)
- relationships: HR = 0.90 (95% CI: 0.88–0.92)

All variables were statistically significant predictors of startup survival.

Higher funding levels were associated with lower failure risk. Similarly, startups completing additional funding rounds exhibited reduced hazard rates. Relationship networks also appeared to contribute positively to startup survival.

In contrast, milestones demonstrated a positive association with failure risk within the fitted model.



6. DISCUSSION

The findings of this study suggest that financial resources play a substantial role in startup survival. Both Kaplan–Meier analysis and Cox regression indicated that higher funding levels are associated with lower failure risk. Access to capital may provide startups with greater flexibility to invest in product development, talent acquisition, and market expansion.

The number of funding rounds also emerged as an important factor. Startups that successfully obtained multiple rounds of investment generally demonstrated improved survival outcomes. This may reflect investor confidence, business scalability, or continued operational performance.

Relationship networks were found to be associated with reduced hazard rates. Stronger connections within entrepreneurial ecosystems may facilitate access to resources, partnerships, expertise, and future investment opportunities.

Industry differences observed in the Kaplan–Meier and log-rank analyses indicate that startup survival is not uniform across sectors. Market conditions, technological uncertainty, regulatory environments, and capital requirements may contribute to varying survival experiences among industries.

The positive association between milestones and failure risk is noteworthy. While milestones are often viewed as indicators of progress, the observed relationship suggests that their role may be more complex within the analysed dataset. Further investigation would be required to better understand this finding.

Overall, the study demonstrates the usefulness of survival analysis techniques in identifying factors associated with startup longevity.

7. *LIMITATIONS*

Several limitations should be considered when interpreting the results.

First, the study relied on observational data obtained from a secondary source. As a result, causal relationships cannot be established.

Second, only variables available within the Crunchbase dataset were analysed. Other potentially relevant determinants of startup survival were not included.

Third, complete-case analysis was used for Cox regression modelling. Excluding observations with missing predictor values reduced the sample size used for regression analysis.

Fourth, proportional hazards testing identified minor violations for log_funding, funding_rounds, and relationships. Although the Cox model was retained due to the exploratory nature of the study and the large sample size, these deviations should be acknowledged when interpreting hazard ratio estimates.

Finally, startups classified as active, acquired, or IPO were treated as censored observations. Alternative event definitions could potentially produce different survival estimates.

8. *CONCLUSION*

This study examined factors affecting startup survival using survival analysis methods applied to venture-funded startups from the Crunchbase Startup Dataset.

The final analytical dataset consisted of 21,725 venture-funded startups, including 1,301 failure events and 20,424 censored observations. Kaplan–Meier survival analysis indicated generally high survival probabilities, while subgroup analyses demonstrated differences across funding levels, funding rounds, and industry categories.

Log-rank tests confirmed statistically significant differences in survival distributions among these groups. Cox proportional hazards modelling further identified funding levels, funding rounds, milestones, and relationship networks as significant predictors of startup survival.

The results suggest that access to funding and stronger relationship networks are associated with reduced startup failure risk. Industry-specific factors also appear to influence survival outcomes.

The study highlights the value of survival analysis techniques for examining entrepreneurial outcomes and contributes to understanding the factors associated with startup longevity in venture-funded environments.

REFERENCES

Crunchbase Startup Dataset. Kaggle. Retrieved from <https://www.kaggle.com/datasets/amirataha/startups>

Hosmer, D. W., Lemeshow, S., & May, S. Applied Survival Analysis: Regression Modeling of Time-to-Event Data.

Kleinbaum, D. G., & Klein, M. Survival Analysis: A Self-Learning Text.

lifelines Documentation. Python Survival Analysis Library.